



Examination Number:

Set:

Shore

Year 12

HSC Assessment Task 4

Half-Yearly Exam

May 2013

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided separately
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–14 show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–14 in a new writing booklet
- Write your examination number on the front cover of each booklet

If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total marks – 70

Section I Pages 3–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 The point P divides the interval from $A(-2, 1)$ to $B(4, -5)$ internally in the ratio $4 : 3$.

What is the x co-ordinate of P ?

(A) $\frac{-17}{7}$

(B) $\frac{10}{7}$

(C) 10

(D) $\frac{10}{12}$

- 2 Simplify $2 \log_x k - \log_x 3 + \log_x p$.

(A) $\log_x \frac{pk^2}{3}$

(B) $\frac{2 \log_x pk}{\log_x 3}$

(C) $\frac{2 \log_x k}{\log_x 3} \times \log_x p$

(D) $\frac{\log_x pk^2}{\log_x 3}$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

3 What is the exact value of $\int_0^{\ln 2} \frac{e^x}{1+e^x} dx$?

- (A) $\ln 2$
- (B) 0
- (C) $e^{\ln 2} - 1$
- (D) $\ln 1.5$

4 What is the exact value of $\int_{-3}^3 \frac{dx}{x^2 + 9}$?

- (A) 0
- (B) $\frac{\pi}{6}$
- (C) $\frac{2}{3}$
- (D) $2\ln 18$

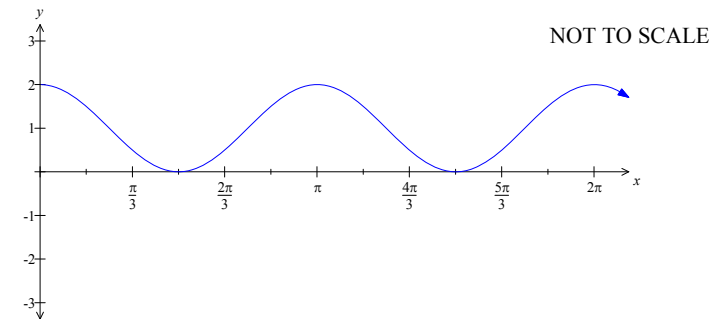
5 The curves $y = x^2$ and $y = x^3$ intersect at the point (1, 1). Which of the following is closest to the size of the acute angle in radians between these curves at (1, 1)?

- (A) 0
- (B) 0.14
- (C) 8.13
- (D) $\frac{\pi}{4}$

6 In a circle a chord of length 10 units is drawn $\sqrt{3}$ units from its centre O . What is the diameter of this circle?

- (A) $2\sqrt{103}$
- (B) 28
- (C) $4\sqrt{7}$
- (D) $\sqrt{28}$

7 Which of the following equations best represents the graph below?



- (A) $y = \sin 2x + 1$
- (B) $y = 1 + 2\cos x$
- (C) $y = 2\cos 2x$
- (D) $y = 1 + \cos 2x$

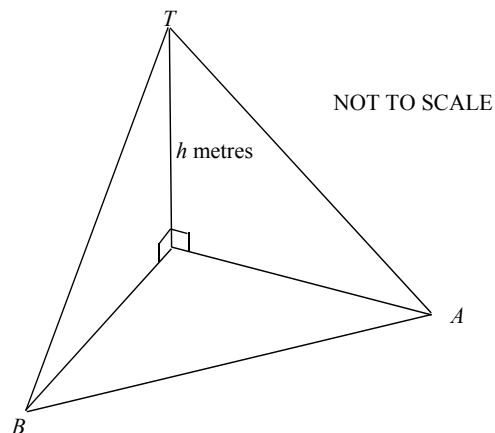
8 Which one of these functions has an inverse relation that is not a function?

- (A) $y = x^3$
- (B) $y = \ln x$
- (C) $y = \sqrt{x}$
- (D) $y = |x|$

9 Given $\frac{d}{dx}(x \log_e x) = 1 + \log_e x$. Find $\int \frac{1 + \log_e x}{x \log_e x} dx$.

- (A) $\frac{1}{x} + c$
 (B) $\log_e(1 + \log_e x) + c$
 (C) $\log_e(x \log_e x) + c$
 (D) $(x \log_e x) + c$

- 10 A tower T , h metres high can be seen from a point A due East of the tower and point B due south of the tower. If the distance between A and B is 60 metres and the angles of elevation of T from A and B are 14° and 17° respectively, find the height of the tower.



- (A) 11.6
 (B) 134.4
 (C) 12.2
 (D) 5.5

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Find the value of k if $x + 3$ is a factor of $P(x) = x^3 - 3kx + 1$. 1
- (b) Solve the inequality $\frac{3x}{2-x} \geq -1$. 3
- (c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{3x}$. 2
- (d) Find $\frac{d}{dx}(x \tan^{-1} 3x)$. 2
- (e) Find $\int \frac{1}{\sqrt{25 - 9x^2}} dx$. 2
- (f) Find the exact value of $\sin\left(\cos^{-1} \frac{2}{7}\right)$. 2
- (g) If α, β, γ are the roots of $2x^3 - 5x^2 + 3x - 1 = 0$, find the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$. 3

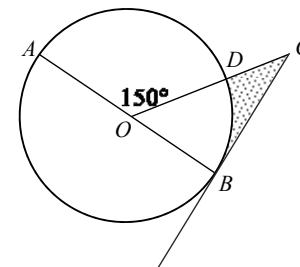
Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the function $y = 2\cos^{-1}\frac{x}{3}$.
- (i) State the domain and range. 2
- (ii) Sketch the curve. 1
- (iii) Find the area between the curve and the x and y axes in the first quadrant. 3
- (b) Use the substitution $u = x^2 + 1$ to evaluate $\int_0^1 x^3(x^2 + 1)^3 dx$. 3
- (c) Consider the function $f(x) = \frac{x}{x+3}$.
- (i) Show that $f'(x) > 0$ for all x in the domain. 1
- (ii) State the equation of the horizontal asymptote of $y = f(x)$. 1
- (iii) Without using further calculus, sketch the graph of $y = f(x)$. 1
- (iv) Explain why $y = f(x)$ has an inverse function $y = f^{-1}(x)$. 1
- (v) Find the inverse function $y = f^{-1}(x)$. 1
- (vi) State the domain of $y = f^{-1}(x)$. 1

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Solve $2\sin^2 x = \sin x$ for $0 \leq x \leq 2\pi$. 2

- (b) In the diagram, AB is a diameter of the circle, centre at O . The radius OD is extended to meet the tangent BC at C . The length of the diameter is 24 cm and $\angle AOC = 150^\circ$.

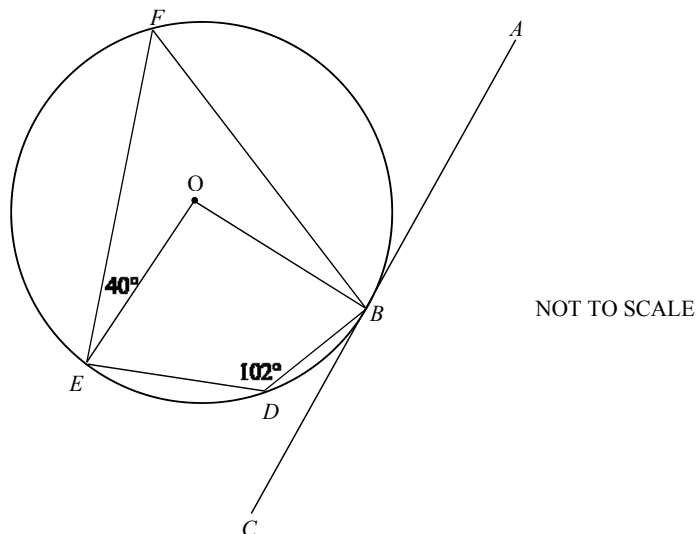


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- (i) Find the size of $\angle COB$ in radians. 1
- (ii) Find the exact value of the shaded area which is contained by the intervals BC and CD and the arc BD . 2
- (c) Consider the function $f(x) = \frac{\ln x}{x}$. 3
- Find the co-ordinates of the stationary point on the curve $y = f(x)$ and determine its nature.
- (d) Use mathematical induction to prove that for all integers $n \geq 1$, $3^{2n-1} + 5$ is divisible by 8. 3

Question 13 continued on the next page

(e)



The points B, D, E, F lie on the circle with centre O . AC is a tangent to the circle touching at the point B .

(i) Show that $\angle BOE = 156^\circ$.

2

(ii) Find the size of the angle FBA .

2

Question 14 on the next page

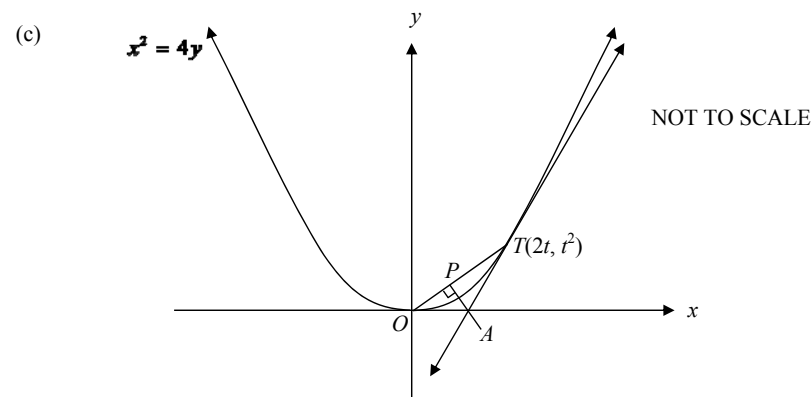
Question 14 (15 marks) Use a SEPARATE writing booklet

(a) Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = 3 \sin x$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x -axis.

3

(b) Prove that $\frac{\sin 2x}{1 + \cos 2x} = \tan x$.

2



The tangent at $T(2t, t^2)$, $t \neq 0$ on the parabola $x^2 = 4y$ meets the x -axis at A . $P(x, y)$ is the foot of the perpendicular from A to OT , where O is the origin. The equation of the tangent at T is $y = tx - t^2$.

(i) Find the co-ordinates of the point A .

1

(ii) Show that the equation of AP is $y = -\frac{2}{t}(x - t)$.

2

(iii) Show that the equation of OT is $t = \frac{2y}{x}$.

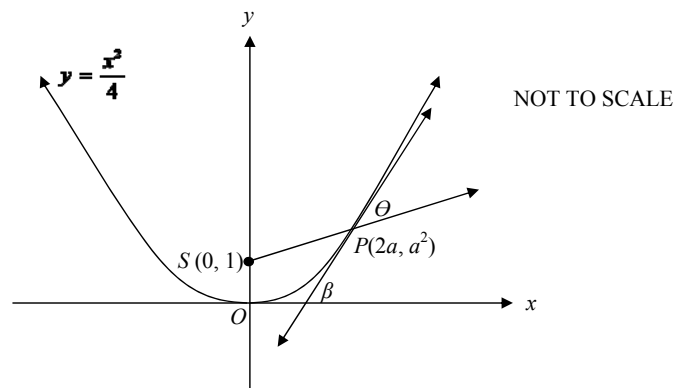
1

(iv) Hence, or otherwise, prove that the locus of $P(x, y)$ lies on a circle with centre $(0, 1)$ and give its radius.

2

Question 14 continued on the next page

(d)



Let $P(2a, a^2)$ be a point on the parabola $y = \frac{x^2}{4}$, and let S be the point $(0, 1)$. The tangent to the parabola at P makes an angle of β with the x axis. The angle between SP and the tangent is θ . Assume $a > 0$, as indicated.

(i) Show that $\tan \beta = a$. 1

(ii) Show that the gradient of SP is $\frac{1}{2}\left(a - \frac{1}{a}\right)$. 1

(iii) Show that $\tan \theta = \frac{1}{a}$. 2

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x$, $x > 0$

YEAR 12 EXTENSION 1 MATHS HALF-YEARLY 2013

1	B	4. $\int_{-3}^3 \frac{dx}{x^2+9} = \frac{1}{3} [\tan^{-1} \frac{x}{3}]_{-3}^3$ $= \frac{1}{3} [\tan^{-1} \frac{3}{3} - \tan^{-1} (-\frac{3}{3})]$ $= \frac{1}{3} [\tan^{-1} 1 - \tan^{-1} (-1)]$ $= \frac{1}{3} [\frac{\pi}{4} + \frac{\pi}{4}]$ $= \frac{\pi}{6}$ (B)
2	A	
3	D	
4	B	
5	B	
6	C	
7	D	
8	D	
9	C	
10	A	

1. A(-2,1) B(4,-5)

4:3

$$x = \frac{4 \times 4 + 3 \times -2}{7}$$

$$= \frac{10}{7} \text{ (B)}$$

2. $\log k^2 - \log 3 + \log p$
 $= \log(\frac{pk^2}{3}) \text{ (A)}$

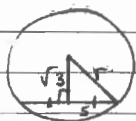
3. $\int_0^{\ln 2} \frac{e^x}{1+e^x} dx = [\ln(1+e^x)]_0^{\ln 2}$
 $= \ln(1+e^{\ln 2}) - \ln(1+e^0)$
 $= \ln 3 - \ln 2$
 $= \ln \frac{3}{2}$
 $= \ln 1.5 \text{ (D)}$

($e^{\ln 2} = 2$)

5. $y = x^2$ $y = x^3$
 $y' = 2x$ $y' = 3x^2$
At $x=1$ At $x=1$
 $y' = 2$ $y' = 3$
 $m_1 = 2$ $m_2 = 3$

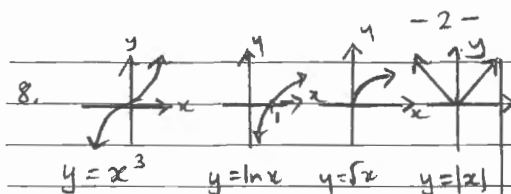
$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$
 $= \frac{2-3}{1+6}$
 $= \frac{1}{7}$
 $\theta = 0.14 \text{ (B)}$

6. $r^2 = (\sqrt{3})^2 + 5^2$
 $= 28$
 $r = \sqrt{28}$
 $= 2\sqrt{7}$
 $\therefore d = 4\sqrt{7} \text{ (C)}$



7. $a = 1$ $P = \frac{5\pi}{6}$
 $b = \frac{b}{\pi} = 2$

$y = 1 + \cos 2x \text{ (D)}$

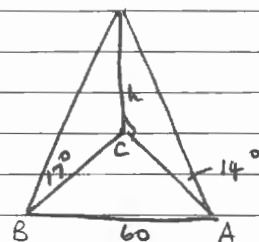


Question 11

8. a) $P(x) = x^3 - 3kx + 1$
 $P(-3) = (-3)^3 - 3(-3)k + 1$
 $0 = -27 + 9k + 1$
 $26 = 9k$
 $k = 2\frac{8}{9} \text{ [1]}$

9. $\int \frac{1+\ln x}{x \ln x} dx$ $[\int \frac{f'(x)}{f(x)}]$
 $= \ln(x \ln x) + C \text{ [C]}$

10.



$\tan 14^\circ = \frac{h}{AC}$
 $AC = \frac{h}{\tan 14^\circ}$
 $BC = \frac{h}{\tan 17^\circ}$
In $\triangle ABC$ Pythagoras' theorem
 $60^2 = \frac{h^2}{\tan^2 14^\circ} + \frac{h^2}{\tan^2 17^\circ}$
 $60^2 \div (\frac{1}{\tan^2 14^\circ} + \frac{1}{\tan^2 17^\circ}) = h^2$
 $\sqrt{60^2 \div 26.78} = h$
 $h = 11.6 \text{ [A]}$

b) $\frac{3x}{2-x} \geq -1$ $[3]$
1. $x \neq 2$
2. Solve $\frac{3x}{2-x} < -1$
 $3x < -2+x$
 $2x < -2$
 $x < -1$

3. $-1 \leq x < 2$

c) $\lim_{x \rightarrow 0} \frac{\sin 4x}{3x} = \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \cdot \frac{4}{3}$
 $= 1 \times \frac{4}{3}$
 $= \frac{4}{3} \text{ [2]}$

d) $\frac{d}{dx} (x \tan^{-1} 3x) = x \times \frac{3}{1+9x^2} + \tan^{-1} 3x$
 $= \frac{3x}{1+9x^2} + \tan^{-1} 3x \text{ [2]}$

$u = x$ $v = \tan^{-1} 3x$
 $u' = 1$ $v' = \frac{3}{1+9x^2}$

-3-

Question 12

$$e) \int \frac{1}{\sqrt{25-9x^2}} dx = \int \frac{1}{\sqrt{9(\frac{25}{9}-x^2)}} dx$$

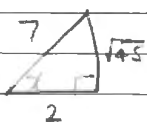
$$= \frac{1}{3} \sin^{-1} \frac{x}{5/3} + C$$

$$= \frac{1}{3} \sin^{-1} \frac{3x}{5} + C \quad [2]$$

$$f) \sin(\cos^{-1} \frac{2}{7}) = \sin x$$

$$\text{let } x = \cos^{-1} \frac{2}{7}$$

$$\cos x = \frac{2}{7}$$



$$\therefore \sin x = \frac{\sqrt{45}}{7}$$

$$= \frac{3\sqrt{5}}{7}$$

[2]

$$g) x^2 y + x^2 y + x^2 y^2 \quad [3]$$

$$= x^2 y (x + y)$$

$$= -\frac{a}{a} x - \frac{b}{a}$$

$$\text{for } 2x^3 - 5x^2 + 3x - 1 = 0$$

$$2 + p + r = \frac{5}{2} \quad x^2 y = \frac{1}{2}$$

$$\therefore x^2 y (x + p + r) = \frac{5}{2} \times \frac{1}{2}$$

$$= \frac{5}{4}$$

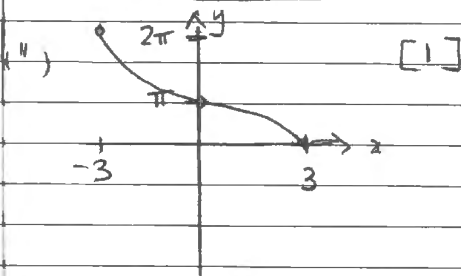
$$(a) -1 < \frac{2}{3} < 1 \quad [2]$$

Domain:

$$-2 < x < 3$$

$$\text{Range: } 0 \leq \frac{y}{2} \leq \pi$$

$$0 \leq y < 2\pi$$



[1]

$$(iii) A = \int 2 \cos^{-1} \frac{x}{3} dx \quad [3]$$

$$y = 2 \cos^{-1} \frac{x}{3}$$

$$\frac{y}{2} = \cos^{-1} \frac{x}{3}$$

$$\cos \frac{y}{2} = \frac{x}{3}$$

$$3 \cos \frac{y}{2} = x$$

$$A = \int 3 \cos \frac{y}{2} dy$$

$$= \left[3 \sin y \right]_0^\pi$$

$$= 6 \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$= 6(1-0)$$

$$= 6u^2$$

-4-

$$(b) \int_0^1 x^3 (x^2+1)^3 dx = \int_0^1 x^2 (x^2+1)^3 x dx \quad [3]$$

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x$$

$$\frac{du}{2} = x dx$$

$$x=1 \quad u=2$$

$$x=0 \quad u=1$$

$$x^2 = u-1$$

$$= \int_1^2 (u-1) u^3 \frac{du}{2}$$

$$= \frac{1}{2} \int_1^2 (u^4 - u^3) du$$

$$= \frac{1}{2} \left[\frac{u^5}{5} - \frac{u^4}{4} \right]_1^2$$

$$= \frac{1}{2} \left[\left(\frac{2^5}{5} - \frac{2^4}{4} \right) - \left(\frac{1}{5} - \frac{1}{4} \right) \right]$$

$$= \frac{49}{40}$$

$$= 1 \frac{9}{40}$$

$$(c) f(x) = \frac{x}{x+3} \quad x \neq -3$$

$$(i) f'(x) = \frac{(x+3)x' - x \cdot 1}{(x+3)^2}$$

$$= \frac{x+3-x}{(x+3)^2}$$

$$= \frac{3}{(x+3)^2}$$

$$(ii) \text{horizontal asymptote}$$

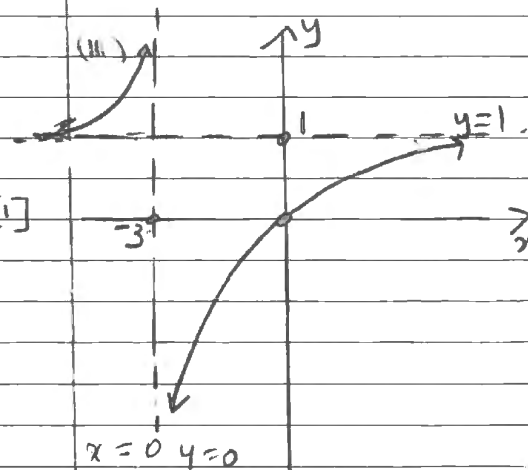
$$y = \frac{x}{x} = \frac{1}{1 + \frac{3}{x}}$$

$$\text{As } x \rightarrow \infty \quad y \rightarrow 1$$

$$\therefore y=1 \quad [1]$$

$$\text{Since } (x+3)^2 > 0 \text{ for all } x$$

$$f'(x) > 0 \text{ for all } x \quad [1]$$



Question 14

a) $V = \pi \int y^2 dx$ [3]
 $= \pi \int_0^{\pi} 9 \sin^2 x dx$
 $= 9\pi \int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2} \cos 2x\right) dx$
 $= 9\pi \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi}$
 $= 9\pi \left[\left(\frac{\pi}{2} - \frac{1}{2} \sin \pi\right) - (0-0) \right]$
 $= 9\pi \left[\frac{\pi}{2} - 0 \right]$

(c)(i) $y = t^2 - t^3$
 let $y=0$ $t^2 = t^3$
 $t = x$
 $\therefore A(t, 0)$ [1]

(ii) $m_{OT} = \frac{t^2-0}{2t-0}$ [2]
 $= \frac{t}{2}$
 $\therefore m_{AP} = -\frac{2}{t}$ (as $OT \perp PA$)

$= \frac{9\pi^2}{4} \cdot \frac{1}{4}$
 $y = 0 = -\frac{2}{t}(x-t)$
 $\therefore y = -\frac{2}{t}(x-t)$ (i)

b) LHS = $2 \sin x \cos x$ [2]

$1 + \cos^2 x - \sin^2 x$

$= \frac{2 \sin x \cos x}{(1 - \sin^2 x) + \cos^2 x}$

$= \frac{2 \sin x \cos x}{\cos^2 x + \cos^2 x}$

$= \frac{2 \sin x \cos x}{2 \cos^2 x}$

$= \frac{\sin x}{\cos x}$

$= \tan x$

$= \text{RHS}$

(iii) Eqn OT [1]

$y - t^2 = \frac{t}{2}(x - 2t)$

$2y - 2t^2 = tx - 2t^2$

$2y = tx$

$2y = t$

$\frac{2y}{x} = t$

(iv) Pt of intersection $P(x, y)$ [2]

Sub $t = \frac{2y}{x}$ into (i)

$y = -\frac{2}{\frac{2y}{x}} \left(x - \frac{2y}{x}\right)$

$= -\frac{x}{y} \left(\frac{x^2 - 2y}{x}\right)$

$y^2 = -x^2 + 2y$

$x^2 + y^2 - 2y = 0$

$x^2 + y^2 - 2y + 1 = 1$

$x^2 + (y-1)^2 = 1$

which is a circle centre $(0, 1)$ $r = 1$ unit

(d)

[1]

(i) $y = \frac{x^2}{2}$

$\frac{dy}{dx} = \frac{x}{1}$

$= \frac{x}{2}$

$= \frac{a^2+1}{2a}$
 $= \frac{1+a^2}{2}$

$= \frac{a^2+1}{2a} \times \frac{2}{1+a^2}$

$= \frac{1}{a}$

At P $\frac{dy}{dx} = \frac{2a}{2}$

$\frac{dy}{dx} = a$ (i)

$m = \tan p$ (i)

$\therefore \tan p = a$ (ii) + (i)

(ii) $m_{SP} = \frac{y_2 - y_1}{x_2 - x_1}$ [1]

$= \frac{a^2-1}{2a-0}$

$= \frac{a^2-1}{2a}$

$= \frac{1}{2} \left(\frac{a^2-1}{a}\right)$

$= \frac{1}{2} \left(a - \frac{1}{a}\right)$

$= \frac{1}{2} \left(a - \frac{1}{a}\right)$

(iii) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ [2]

$= a - \frac{1}{2} \left(a - \frac{1}{a}\right)$

$= \frac{1 + a \left(a - \frac{1}{a}\right)}{2}$

$= a - \frac{1}{2} a + \frac{1}{2a}$

$= \frac{1 + a^2 - \frac{1}{2}}{2}$

$= \frac{\frac{1}{2} a + \frac{1}{2a}}{2}$

$= \frac{1}{2} + \frac{a^2}{2}$